

## CSE 2<sup>nd</sup> Year KCS302 (Computer Organization) Question Bank (Unitwise)

| Q.No. | Question Description                                                                                                                                                                                                                                                 | Course Outcome | Marks | Difficulty Level |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-------|------------------|
| 1     | Describe the digital computer generation in brief.                                                                                                                                                                                                                   | CO1            | 2     | Low              |
| 2     | Define bus arbitration with suitable diagram.                                                                                                                                                                                                                        | CO1            | 5     | Medium           |
| 3     | Explain different functions of a computer system.                                                                                                                                                                                                                    | CO1            | 5     | Medium           |
| 4     | Write short note on the following:<br>(i) Common Bus system<br>(ii) Bus Arbitration                                                                                                                                                                                  | CO1            | 5     | Medium           |
| 5     | Draw a diagram of a bus system for four registers that uses three state buffer and a decoder instead of the multiplexers                                                                                                                                             | CO1            | 5     | High             |
| 6     | Explain ALU (Arithmetic Logic Unit) ? Draw logic diagram of ALU that performs AND, OR logic operations and ADD, SUB arithmetic operations                                                                                                                            | CO1            | 10    | Medium           |
| 7     | Compute the following arithmetic operations in 8-bit registers. Use signed 2's complement notation. Indicate overflow/underflow, if any :—<br>(a) - 28 - (- 100)<br>(b) - 28 - 100<br>(c) 78 - (- 49)<br>(d) +50-5                                                   | CO1            | 10    | Medium           |
| 8     | Elaborate the following conditional control statement by two register transfer statements with control functions. If (P = 1) then (R1 ← R2) else if (Q = 1) then (R1 ∧ R3) 11. A computer has 16 registers, an ALU with                                              | CO1            | 5     | Medium           |
| 9     | Explain various addressing modes with suitable examples.                                                                                                                                                                                                             | CO1            | 10    | Medium           |
| 10    | Describe stack organization ? Compare Register stack and Memory stack.                                                                                                                                                                                               | CO1            | 10    | Medium           |
| 11    | Describe use of registers in computer? Explain general Register organization with multiple bus                                                                                                                                                                       | CO1            | 5     | High             |
| 12    | Explain the need of having many addressing modes in your machine ? Discuss Indirect and displacement addressing in detail.                                                                                                                                           | CO1            | 5     | Medium           |
| 13    | The following statements specify a memory. Explain the memory operation in each case<br>(a) $R2 \leftarrow M[AR]$<br>(b) $M[AR] \leftarrow R3$<br>(c) $R5 \leftarrow M[R5]$                                                                                          | CO1            | 5     | Medium           |
| 14    | An 8-bit register contains the binary value 10011100. Calculate the register value after an arithmetic shift right? Starting from the initial number 10011100, determine the register value after the arithmetic shift left and state whether, there is an overflow. | CO1            | 2     | Medium           |

## CSE 2<sup>nd</sup> Year KCS302 (Computer Organization) Question Bank (Unitwise)

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| 15 | Review about wrong with the following register transfer statements? a) $xT:AR \leftarrow AR, AR \leftarrow 0$ b) $yT:R1 \leftarrow R2, R1 \leftarrow R3$                                                                                                                                                                                                                                          | CO1 | 2  | Medium |
| 16 | Write a program to evaluate arithmetic expressions using three, zero address instruction:<br>$X = (A-B) + C * (D * E - F) / G + H * K$                                                                                                                                                                                                                                                            | CO1 | 10 | High   |
| 17 | The memory unit of a computer has 256 words of 32 bits each .the computer has an instruction format with four fields : an operation code, a mode field to specify one of seven addressing modes, a register address field to specify one of 60 processor registers and a memory address. state the instruction format and the number of bits in each field if the instruction in one memory word. | CO1 | 10 | High   |
| 18 | An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contains the number 200.Evaluate the effective address if the addressing mode of the instruction is a) direct b)immediate c) relative d) register indirect e) index with R1 as the index register.                                                  | CO1 | 10 | Medium |
| 19 | Evaluate the no. of address and data line in a main memory of size 64K*16                                                                                                                                                                                                                                                                                                                         | CO1 | 2  | Low    |
| 20 | A digital computer has a common bus system for 32 registers of 16 bits each. The bus is constructed with multiplexers. How many selection inputs are there in each multiplexer? How many multiplexers are there in the bus? What is the size of each MUX.                                                                                                                                         | CO1 | 5  | Medium |
| 21 | Explain Booth's Algorithm in detail with flow chart. Implement multiplication of two operands (-12) and (+4) with the help of Booth's algorithm.                                                                                                                                                                                                                                                  | CO2 | 10 | High   |
| 22 | Explain Unsigned multiplication algorithm with flow chart. Implement multiplication of 12 and 4 using Unsigned multiplication algorithm                                                                                                                                                                                                                                                           | CO2 | 10 | Medium |
| 23 | Design a 3*3 bit array multiplier with a suitable diagram                                                                                                                                                                                                                                                                                                                                         | CO2 | 5  | Medium |
| 24 | Discuss IEEE standard for floating point numbers.                                                                                                                                                                                                                                                                                                                                                 | CO2 | 2  | Medium |
| 25 | Discuss and Differentiate between fixed point representation and floating point representation. Explain with suitable examples                                                                                                                                                                                                                                                                    | CO2 | 5  | Medium |
| 26 | Explain the following with reference to floating point representation: —<br>(i)IEEE standard<br>(ii) Normalization<br>(iii) Biasing<br>(iv) Overflow & Underflow                                                                                                                                                                                                                                  | CO2 | 10 | Low    |
| 27 | State Booth Algorithm for multiplication of two numbers. Draw a logic diagram for the implementation of the Booth Algorithm for determining the product of two 8-bit signed numbers                                                                                                                                                                                                               | CO2 | 10 | Medium |
| 28 | State the division of the following positive number 1000/11. using 4-bit register.                                                                                                                                                                                                                                                                                                                | CO2 | 10 | High   |
| 29 | Design the carry look ahead adder.                                                                                                                                                                                                                                                                                                                                                                | CO2 | 10 | Medium |
| 30 | Design carry look ahead generator.                                                                                                                                                                                                                                                                                                                                                                | CO2 | 5  | High   |

## CSE 2<sup>nd</sup> Year KCS302 (Computer Organization) Question Bank (Unitwise)

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| 31 | Explain floating point addition and subtraction algorithm.                                                                                                                                                                                                                                                                                                                                                | CO2 | 10 | High   |
| 32 | Define Floating point overflow and underflow conditions.                                                                                                                                                                                                                                                                                                                                                  | CO2 | 2  | Medium |
| 33 | Explain the various steps required to design ALU.                                                                                                                                                                                                                                                                                                                                                         | CO2 | 10 | High   |
| 34 | Explain steps to divide two signed numbers.                                                                                                                                                                                                                                                                                                                                                               | CO2 | 5  | Medium |
| 35 | Explain the process of dividing $+7/-3$ using restoring method.                                                                                                                                                                                                                                                                                                                                           | CO2 | 5  | High   |
| 36 | Explain signed multiplication algorithm with the help of flow chart. Implement multiplication of $(+12)*(-4)$ using Unsigned multiplication algorithm                                                                                                                                                                                                                                                     | CO2 | 10 | High   |
| 37 | Design a 4 bit arithmetic circuit using a suitable block diagram.                                                                                                                                                                                                                                                                                                                                         | CO2 | 5  | Medium |
| 38 | Design a 4 bit Bidirectional shifter with the help of multiplexer.                                                                                                                                                                                                                                                                                                                                        | CO2 | 5  | Medium |
| 39 | Explain divide overflow.                                                                                                                                                                                                                                                                                                                                                                                  | CO2 | 2  | Medium |
| 40 | Describe carry look ahead in digital circuits.                                                                                                                                                                                                                                                                                                                                                            | CO2 | 2  | Low    |
| 41 | Define CISC ? Explain it with its characteristics.                                                                                                                                                                                                                                                                                                                                                        | CO3 | 5  | High   |
| 42 | A computer has 16 registers, an ALU with 32 operations, and a shifter with eight operations, all connected to a common bus system .<br>(i) Formulate a Control Word for a micro-operation.<br>(ii) Specify the number of bits in each field of the control word and give a general encoding scheme.<br>(iii) Show the bits of the control word that specify the micro-operation $R4 \leftarrow R5 + R6$ . | CO3 | 10 | High   |
| 43 | Define microoperation? How can microoperation be used for execution of an instruction? Explain with the help of an example.                                                                                                                                                                                                                                                                               | CO3 | 5  | Medium |
| 44 | Instruction cycle is divided into subcycles. With the help of diagram, explain the sequence in which subcycles are executed                                                                                                                                                                                                                                                                               | CO3 | 5  | Medium |
| 45 | Discuss how address sequencing is carried out in microprogrammed organization.                                                                                                                                                                                                                                                                                                                            | CO3 | 5  | High   |
| 46 | Describe how control unit of a computer function? Explain with the help of a block diagram                                                                                                                                                                                                                                                                                                                | CO3 | 10 | Medium |
| 47 | Define the following :<br>(i) Microoperation<br>(ii) Microinstruction<br>(iii) Microprogram<br>(iv) Microcode                                                                                                                                                                                                                                                                                             | CO3 | 2  | Medium |
| 48 | Describe the basic differences between a branch instruction, a call subroutine instruction, and program interrupt?                                                                                                                                                                                                                                                                                        | CO3 | 5  | Medium |
| 49 | Describe the differences between CISC & RISC architecture.                                                                                                                                                                                                                                                                                                                                                | CO3 | 5  | Medium |
| 50 | State the meaning of hard wired control unit? Give various methods to design hardwired control unit. Describe one of the design methods for hardwired control unit with suitable diagrams.                                                                                                                                                                                                                | CO3 | 10 | High   |

## CSE 2<sup>nd</sup> Year KCS302 (Computer Organization) Question Bank (Unitwise)

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| 51 | Explain wide-branch addressing? Explain with example.                                                                                                                                                                                                             | CO3 | 5  | High   |
| 52 | Define Microinstruction ? How is it different from microprogram sequencer ? Explain with the help of example.                                                                                                                                                     | CO3 | 5  | Medium |
| 53 | Describe how a processor execute instructions? Define the internal functional units of a processor and how they are interconnected                                                                                                                                | CO3 | 10 | Medium |
| 54 | A system uses a control memory of 1024 words of 32 bits each..The micro-instruction has three fields:select, address and microoperation fields. The microoperation's field has 16 bits. (a) How many bits are there in the branch address field and select field? | CO3 | 10 | Medium |
| 55 | Using the mapping procedure, determine the first micro-instruction address for the following operation code:<br>a)0010 b)1011 c)1111                                                                                                                              | CO3 | 5  | High   |
| 56 | Describe the difference between a microprocessor and a microprogram? Is it possible to design a microprocessor without a microprogram? Explain.                                                                                                                   | CO3 | 5  | High   |
| 57 | Define interrupt? Explain various types of interrupt.                                                                                                                                                                                                             | CO3 | 5  | High   |
| 58 | Explain the hardwired control unit organization explaining each component clearly                                                                                                                                                                                 | CO3 | 10 | Medium |
| 59 | Explain advantages of pipelining with a suitable example.                                                                                                                                                                                                         | CO3 | 5  | High   |
| 60 | Define program control. Explain its types.                                                                                                                                                                                                                        | CO3 | 5  | Medium |
| 61 | Write short notes on any two of the following :<br>(i) Auxiliary memory<br>(ii) Memory Hierarchy<br>(iii) Virtual Memory (iv) Cache Memory                                                                                                                        | CO4 | 5  | Medium |
| 62 | Define semiconductor RAM memories? Show the read operation and write operation in static memories with examples                                                                                                                                                   | CO4 | 5  | Medium |
| 63 | Explain the concept of Virtual memory. How address mapping is performed in virtual memory ?                                                                                                                                                                       | CO4 | 5  | High   |
| 64 | Define the difference between 2D and 2 1/2 D memory organization ? Explain it with the help of suitable examples                                                                                                                                                  | CO4 | 5  | Medium |
| 65 | A ROM chip of 1024 x 8 bits has four select inputs and operates from a 5-volt power supply. How many pins are needed for the IC package? Draw a block diagram and label all inputs and output terminals in the ROM.                                               | CO4 | 2  | Low    |
| 66 | A two-way set associative cache memory uses blocks of four words. The cache can accommodate a total of 2048 words from the main memory. The main memory size is 128 K*32.<br>(i) Formulate all pertinent information required to construct the cache              | CO4 | 10 | High   |

## CSE 2<sup>nd</sup> Year KCS302 (Computer Organization) Question Bank (Unitwise)

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|    | memory.<br>(ii) What is the size of the cache memory                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |    |        |
| 67 | Define the terms address space and memory space. An address space is specified by 24 bits and the corresponding memory space by 16 bits. Find the following :<br>(i) How many words are there in the address space?<br>(ii) How many words are there in the memory space                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CO4 | 5  | Medium |
| 68 | Discuss the various organization of RAM. A computer uses RAM chips of 1024x1 capacity. How many chips are needed and how should their address lines be connected to provide a memory capacity of 1024 bytes ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO4 | 5  | Medium |
| 69 | Explain various cache mapping techniques. A computer system has a 4k word cache organised in block set associative manner with 4 blocks per set, 64 words per block. The main memory contains 65536 blocks. How many bits are there in each of the TAG, SET & WORD fields ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO4 | 10 | Medium |
| 70 | Describe the difference between Direct mapping and Associative mapping procedures for organisation of cache memory with example. Give merits and demerits of both mapping procedures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO4 | 10 | High   |
| 71 | Evaluate how many 128 bytes RAM chips are required to provide a memory of 2048 bytes ? Show details of connection clearly indicating address, data and decoder configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | CO4 | 5  | High   |
| 72 | A virtual memory system has a page size of 1K words. There are eight pages and four blocks. The associative memory page table contains the following entries:<br><div style="display: flex; justify-content: space-between; width: 100%;"> <span>Page</span> <span>Block</span> </div> <div style="display: flex; justify-content: space-between; width: 100%;"> <span>0</span> <span>3</span> </div> <div style="display: flex; justify-content: space-between; width: 100%;"> <span>1</span> <span>1</span> </div> <div style="display: flex; justify-content: space-between; width: 100%;"> <span>4</span> <span>2</span> </div> <div style="display: flex; justify-content: space-between; width: 100%;"> <span>6</span> <span>0</span> </div> Make a list of all virtual address (in decimal) that will cause a page fault, if used by the processor. | CO4 | 5  | Medium |
| 73 | The logical address space in a computer system consists of 129 segments. Each segment can have up to 32 pages of 4K word in each. Physical memory consists of 4K Blocks of 4K words in each. Formulate the logical and physical address formats.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO4 | 5  | Medium |
| 74 | computer employs RAM chips of 1024*8 and ROM chips of 512*8. The computer system needs 2KB of RAM 4KB of ROM and four interface units, each with four registers. A memory mapped I/O configuration is used. The two highest order bits of the address bus assigned 00 for RAM, 01 for ROM and 10 for interface registers.<br>a). determine RAM and ROM chips are needed?<br>b) .Draw a memory-address map for the system                                                                                                                                                                                                                                                                                                                                                                                                                                   | CO4 | 10 | High   |
| 75 | Describe memory hierarchy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CO4 | 2  | Low    |
| 76 | Explain various types of ROM ?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CO4 | 5  | Medium |

## CSE 2<sup>nd</sup> Year KCS302 (Computer Organization) Question Bank (Unitwise)

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| 77  | Write the difference between SRAM & DRAM?                                                                                                                                 | CO4 | 2  | Low    |
| 78  | Define locality of reference. Explain with an example?                                                                                                                    | CO4 | 5  | Medium |
| 79  | Describe Cache memory. How it is constructed.                                                                                                                             | CO4 | 2  | Medium |
| 80  | Describe why cache memory mapping is best and why?                                                                                                                        | CO4 | 2  | Medium |
| 81  | Explain Direct Memory Access (DMA).                                                                                                                                       | CO5 | 5  | Medium |
| 82  | Explain Synchronous and Asynchronous communication.                                                                                                                       | CO5 | 5  | Medium |
| 83  | Define Interrupts with their types and exceptions                                                                                                                         | CO5 | 5  | High   |
| 84  | Write short note on the following together with their importance:<br>(i) Input-output processor<br>(ii) Serial Communication.                                             | CO5 | 5  | Medium |
| 85  | Describe why Input-Output interface is required? Describe various methods for I/O interface together with their merits and demerits                                       | CO5 | 10 | Medium |
| 86  | Define what programming steps are required to check when a source interrupts the computer while it is still serviced by a previous interrupt request from the same source | CO5 | 5  | High   |
| 87  | Draw a flow chart that describes the CPU-I/O channel communication in the IBM 370.                                                                                        | CO5 | 10 | Medium |
| 88  | Describe what is basic advantage of using interrupt initiated data transfer over transfer under program control without an interrupt ?                                    | CO5 | 5  | Medium |
| 89  | Describe Strobe control and Hand shaking for Asynchronous Data Transfer.                                                                                                  | CO5 | 10 | Medium |
| 90  | Define programmed I/O.                                                                                                                                                    | CO5 | 2  | Medium |
| 91  | Explain various types of peripheral devices with a suitable example.                                                                                                      | CO5 | 5  | Low    |
| 92  | Determine the different modes of data transfer. Explain any one of them with a suitable example.                                                                          | CO5 | 5  | Medium |
| 93  | With the help of block diagram, explain working of Direct Memory Access (DMA). Give a brief comparison of programmed I/O & Interrupt I/O.                                 | CO5 | 10 | Medium |
| 94  | Explain strobe methods with suitable diagram.                                                                                                                             | CO5 | 5  | High   |
| 95  | Explain standard communication interface with a suitable diagram.                                                                                                         | CO5 | 5  | Medium |
| 96  | Describe the advantage of using input output interface ?                                                                                                                  | CO5 | 2  | Medium |
| 97  | Write the difference between I/O Channel and I/O processors.                                                                                                              | CO5 | 5  | Medium |
| 98  | Explain the difference between external and internal interrupt.                                                                                                           | CO5 | 5  | Medium |
| 99  | Explain structure of Magnetic hard disk along with read and write operations.                                                                                             | CO5 | 5  | Low    |
| 100 | Explain the exceptions in interrupt handling process ?.Explain in detail ?.                                                                                               | CO5 | 5  | Medium |